RIEMANNIAN GEOMETRY (MATH 319)

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These are my notes for Topology III taught by Andre Neves at UChicago in Spring 2026. Textbook: *Riemannian Geometry* by Manfredo do Carmo.

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1. Introduction

A *Riemannian Manifold* is a manifold M that is also a metric space with some metric g (with some additional properties). We will see that such a manifold has a "curvature" $K(g)$. The basic questions in this subject are:

(1) What does $K(g)$ tell us about M?
(2) Given M and varying g, what can $K(g)$ be?

An example of the first type of question comes from physics: the Einstein Equation.

The second question is solved in dimensions 2 and 3, which are the results of Riemann Uniformization and the Poincaré Geometrization Conjecture respectively.

1.1. Riemannian Metrics.

More precisely, a Riemannian Manifold must have the following two properties of its metric g:

(i) g comes from a dot product on TM, i.e. we have bilinear maps $g_p : T_pM \times T_pM \to \mathbb{R}$ at each $p \in M$.
(ii) For any chart $\varphi : M \to \mathbb{R}^n$, $g_x(\partial_i\varphi, \partial_j\varphi)$ is smooth in x.

Given such a g, one can define the *length* of curves on M by integrating the length-form over the curve, and similarly we can define the volume of a subset of M.

The important equivalence class for such manifolds is *isometry*. A *local isometry* is a map $F : (M, g) \to (N, h)$ such that $F^*h = g$. It is a *global isometry* if F is a diffeomorphism. "Almost all" metrics have no nontrivial isometries, but they are common in examples we care about.

Examples:

- *Flat*: $M = \mathbb{R}^n$, standard inner product.
- *Spherical*: $M = S^n$, standard inner product induced by inclusion in $\mathbb{R}^{n+1}$.
- *Hyperbolic*: $M = B^n$,
$$h_p(\vec{x}, \vec{y}) = \frac{4}{(1 - |p|^2)^2}\langle \vec{x}, \vec{y}\rangle.$$

The isometries of $\mathbb{R}^n$ and S^n are $\mathrm{O}_n(\mathbb{R})$ + affine transformations and $\mathrm{O}_n(\mathbb{R})$ respectively. For hyperbolic space, the isometries are conformal maps.

If G is a discrete subgroup of the group of isometries of (M, g) which acts on M freely and properly discontinuously, then we have a *quotient manifold*
$$M/G := \frac{M}{\{x \sim \varphi(x) \;\; \varphi \in G\}}.$$

Examples of Quotients:

- Take some $\mathbb{Z}^n$ as a subgroup of affine transformations on $\mathbb{R}^n$, for which $\mathbb{R}^n/\mathbb{Z}^n = T^n$, the n-torus.
- Let $M = S^3$ treated as a subspace of $\mathbb{C}^2$. Let φ be a rotation in each complex component with periods p, q that are *coprime*, and we have
$$L_{p,q} := S^3/\langle\varphi\rangle$$
called the "Lens Space" or sometimes "Poincaré Sphere."

1.2. Connections.

Let $\mathfrak{X}(M)$ be the space of smooth vector fields on M. $\mathfrak{X}(M)$ has the structure of a vector space with coefficients in $C^\infty(M)$. Moreover, there is also an action of $\mathfrak{X}(M)$ on $C^\infty(M)$, given by $Vf := \partial_V f$. The Riemannian metric on M gives us an inner product $\langle\bullet, \bullet\rangle : \mathfrak{X}(M)^2 \to C^\infty(M)$.

A *connection* is a map $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ satisfying

- Linearity in the first argument: $\nabla_{f\cdot X}Y = f \cdot \nabla_X Y$.
- $\nabla_X(f \cdot Y) = \partial_X f \cdot Y + f \cdot \nabla_X Y$ (Leibniz rule)

One should think of $\nabla_X Y$ as the "derivative of Y as it flows along X." Connections allow us to assign a derivative for vector fields along curves: if V is a vector field defined on the curve $c : I \to M$, the "covariant derivative" is another vector field on c denoted $DV/\mathrm{d}t$ and defined

$$\frac{DV}{\mathrm{d}t}(t_0) = (\nabla_{(\mathrm{d}c/\mathrm{d}t)^{ext}} V^{ext})(c(t_0))$$

where $(\mathrm{d}c/\mathrm{d}t)^{ext}$ and V^{ext} are extensions of $\mathrm{d}c/\mathrm{d}t, V$ to a neighborhood of $c(t_0)$.

Lemma (Properties of covariant derivative): $DV/\mathrm{d}t$ is the unique vector field satisfying the following:

(a) Linearity in V: $D/\mathrm{d}t(V + W) = DV/\mathrm{d}t + DW/\mathrm{d}t$.
(b) Leibniz rule: $D/\mathrm{d}t(f \cdot V) = \mathrm{d}f/\mathrm{d}t \cdot V + f \cdot DV/\mathrm{d}t$.
(c) If V is defined on all of M, then $DV/\mathrm{d}t = \nabla_{\mathrm{d}c/\mathrm{d}t} V$.

Proof. do later □

Given a connection ∇, one can define the *torsion* and *curvature*

- $T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y]$
- $R(X, Y)Z := (\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]})(Z)$

both of which are $C^\infty(M)$-linear in all arguments (unlike $\nabla_X Y$ itself). For example,

$$\begin{aligned} T(fX, Y) &= \nabla_{fX} Y - \nabla_Y (fX) - [fX, Y] \\ &= f \cdot \nabla_X Y - (\partial_Y f \cdot X + f \cdot \nabla_Y X) - (\partial_Y f \cdot X + f \cdot [X, Y]) \\ &= f \cdot \nabla_X Y - f \cdot \nabla_Y X - f \cdot [X, Y] \\ &= f \cdot T(X, Y). \end{aligned}$$

In general there are many connections ∇ on any given Riemannian manifold, so it would be nice to have some canonical choice. And there is one ∇ that is especially good:

Theorem (Levi-Civita): Given (M, g), there is a unique connection ∇ which is

- *Symmetric*, meaning $\nabla_X Y - \nabla_Y X = [X, Y]$ (equivalently $T(X, Y) = 0$ for all X, Y).
- *Compatible* with g, meaning $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$ (the product rule).

Such a ∇ is called a *Levi-Civita* or *Riemannian* connection.

Proof. We can show with some algebra that the two conditions are equivalent to

$$\langle Z, \nabla_Y X\rangle = \frac{1}{2}\Big(X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle [X, Z], Y\rangle - \langle [Y, Z], X\rangle - \langle [X, Y], Z\rangle\Big)$$

so the unique connection is both given and determined by this equation, called the *Koszul formula.* □

From now on we'll assume that ∇ is the Levi-Civita connection.

To actually compute $\nabla_A B$ at $p \in M$ for two given vector fields $A, B \in \mathfrak{X}(M)$, we may want to choose local coordinates in a neighborhood U of p. If the local coordinates are $x^1, \ldots, x^n$ and the vector fields at p are given the basis $\partial_1, \ldots, \partial_n$, then we can compute $(\nabla_A B)_p$ by writing $A = \sum_i a_i \partial_i$ and $B = \sum_i b_i \partial_i$ $(a_i, b_i \in C^\infty(U))$ and using the properties of ∇:

$$\begin{aligned} \nabla_A B &= \sum_i a_i \nabla_{\partial_i} B \\ &= \sum_i a_i \sum_j b_j \nabla_{\partial_i} \partial_j + \partial_i b_j \cdot \partial_j \\ &= \Big(\sum_{ij} a_i b_j \nabla_{\partial_i} \partial_j\Big) + \Big(\sum_{ij} a_i \partial_i b_j \cdot \partial_j\Big). \end{aligned}$$

These functions $\nabla_{\partial_i}\partial_j$ are important, so we abbreviate

$$\nabla_{\partial_i}\partial_j =: \sum_k \Gamma_{ij}^k \partial_k.$$

The coefficient function $\Gamma_{ij}^k \in C^\infty(U)$ is called the *Christoffel Symbol*, and it depends only on g. Note that if ∇ is the Levi-Civita connection, then $\Gamma_{ij}^k = \Gamma_{ji}^k$, as $[\partial_i, \partial_j] = 0$. In terms of Γ_{ij}^k we have

$$\nabla_A B = \sum_k \left(\sum_{ij} a_i b_j \Gamma_{ij}^k + \partial_A b_k \right) \cdot \partial_k.$$

This shows that $(\nabla_A B)_p$ depends only on $A(p), B(p)$, and $\partial_{A(p)} b_k$. In particular, $\nabla_A B$ depends on $A(p)$ but not any other values of A, whereas for B the local behavior near p matters too.

In the specific case of $\mathbb{R}^n$ with the Euclidean metric, $\Gamma_{ij}^k = 0$, so this becomes

$$\nabla_A^{\mathbb{R}^n} B = \sum_k \partial_A b_k \partial_k.$$

Another notable example is on S^n, where the Levi-Civita connection is

$$\nabla_A^{S^n} B = (\nabla_{A^{ext}}^{\mathbb{R}^{n+1}} B^{ext})^T$$

where A^{ext}, B^{ext} are A, B extended to $\mathbb{R}^{n+1}$ and X^T is the orthogonal projection of $X(p)$ onto $T_p S^n$.

To compute the Christoffel symbols themselves, we have (following from the Levi-Civita theorem)

$$\Gamma_{ij}^k = \frac{1}{2} \sum_\ell \Big(\partial_i g_{j\ell} + \partial_j g_{\ell i} - \partial_\ell g_{ij} \Big) g^{\ell k}$$

where $g_{ij} := \langle \partial_i, \partial_j \rangle$, a smooth function on U for each i, j, and g^{ij} is defined as the i, j entry in the inverse matrix $(g^{ij}) = (g_{ij})^{-1}$.

1.3. **Geodesics and the Exponential Map.** In a Riemannian manifold, given a curve $\gamma : I \to M$ we have a γ' and $\gamma'' = \nabla_{\gamma'}\gamma'$ given by the Levi-Civita connection. γ is called a *geodesic* when $\gamma'' = 0$. This implies that γ can be parameterized by arc length, as $|\gamma'|$ is constant.

Another slightly broader definition of geodesic is a curve $\gamma : I \to M$ for which

$$\gamma''(t) = \lambda(t)\gamma'(y)$$

which means, equivalently, that γ can be re-parameterized such that $\gamma''(t) = 0$.

Examples:

- Let $M = S^n$. The geodesics are curves contained in the great circles. For these curves $\gamma'' = -\gamma$. There is an acceleration but it is "invisible to the sphere" because it is orthogonal to the tangent space.
- Let $M = B^n$ with the hyperbolic metric. A straight line away from the origin can be re-parameterized to be a geodesic, by taking the distance to be $\tanh(t)$. Away from the origin, the geodesics are parts of circles meeting the boundary orthogonally.

Proposition: In general if γ is a curve on M and there is an isometry $P : M \to M$ for which $P(\gamma(t)) = \gamma(t)$ but P reverses orientation of normal vectors to γ at each point, then γ is a geodesic.

On hyperbolic space, the isometry corresponding to each geodesic is like an involution.

Theorem (Existence and Uniqueness of Geodesics): For each $p \in M$ and $v \in T_p M$, locally there is a unique curve $\gamma : (-\varepsilon, \varepsilon) \to M$ which is a geodesic, $\gamma(0) = p$, and $\gamma'(0) = v$. We call this $\gamma(t) =: \exp_x(tv)$.

Proof. If $\varphi : U \to \mathbb{R}^n$ is a chart near p with local coordinates $x_1, \ldots, x_n$, then the condition $D\gamma'/\mathrm{d}t = 0$ just becomes a second-order ODE involving Γ_{ij}^k:

$$x_k'' + \sum_{ij} x_i' x_j' \Gamma_{ij}^k = 0 \quad \forall 1 \leq k \leq n$$

with initial conditions on $\gamma(0)$ and $\gamma'(0)$, so there is a unique solution. □

The map $\exp_x$ is very nice. We'll now show some of its properties.

Gauss's Lemma: Fixing $p \in M$, for all $v, w \in T_pM$ with $|v| < \varepsilon$,

$$\langle (\mathrm{d}\exp)_v(v), (\mathrm{d}\exp)_v(w) \rangle = \langle v, w \rangle$$

In other words, $\exp_p$ is locally a conformal map.

Proof. By linearity we can break into cases of orthogonal and parallel. If $w = \lambda v$ then letting $\gamma(t) = \exp(tv)$,

$$\mathrm{d}\exp(\lambda v) = \lambda \gamma'(1),$$

so we have

$$\langle \mathrm{d}\exp(v), \mathrm{d}\exp(w) \rangle = \lambda \langle \gamma'(1), \gamma'(1) \rangle = \lambda |v|^2 = \langle v, w \rangle$$

On the other hand, if $w \perp v$, and wlog $|w| = |v|$, then let $w : (-\varepsilon, \varepsilon) \to T_pM$ be the circular curve with $w(0) = v$, $w'(0) = w$, and $|w(t)| = |v|$ for all t. Now let $F : (\varepsilon, \varepsilon) \times (0, 1) \to M$ be defined

$$F(s, t) := \exp_p(t \cdot w(s)).$$

For each fixed s, $F(s, \bullet)$ is a geodesic curve in M. *fill in details later!!!* □

As geodesics sprout from x in all directions, it may be that some of them intersect, and it may be that they don't. As it turns out, they always go a non-zero distance before intersecting.

Proposition (Normal Neighborhood): For $p \in M$, there is a neighborhood U and $\varepsilon < r$ (for some r called the *injectivity radius* at x) for which:

(i) $\exp_q : B_\varepsilon(U) \to M$ is a diffeomorphism onto its image.
(ii) For all $q_1, q_2 \in U$, there is a geodesic connecting q_1 and q_2 (not necessarily contained in U).

Proof. Fix a local chart $\varphi : \tilde{U} \to M$ near p so that $\varphi(0) = p$. By the previous theorem, for some $V - 1 \subset \tilde{U}$ and $\varepsilon_1 > 0$ we have a well-defined map $F : V_1 \times B_{\varepsilon_1} \to M \times M$ where

$$F(x, v) = (\varphi(x), \exp_{\varphi(x)}(v)).$$

Now we just need to choose V_1 so that F is a diffeomorphism onto its image, for which we can use the implicit function theorem. It suffices to show that $\mathrm{d}F$ is bijective at 0. And we can compute, using the definition of geodesics, that at $(0, 0)$,

$$\mathrm{d}F = \begin{pmatrix} \mathrm{d}\varphi & \mathrm{d}\varphi \\ 0 & \mathrm{d}\varphi \end{pmatrix}$$

which is invertible. □

The shortest path between any two points on a closed manifold is always a geodesic, though there are many geodesics of different lengths.

Proposition (Normal Neighborhood fact): Let U_x be a normal neighborhood containing point x with corresponding value ε_x, and let $q \in U_x$. Then

(i) There is a unique geodesic passing through x, q with length less than ε_x.
(ii) This geodesic minimizes distance among all smooth curves from x to q.

Proof. (i): Let $\exp(tw)$ and $\exp(tv)$ be two different geodesics going to q. The lengths of these geodesics are $|w|$ and $|v|$, as

$$\int_0^1 |\partial_t \exp(tw)| \, \mathrm{d}t = \int_0^1 |w| = |w|$$

because geodesics have constant velocity. If both $|v|$ and $|w|$ are less than ε_x, then exp is not injective, contradicting the normal neighborhood. And if neither are, then exp is not surjective.

(ii): Suppose we have a curve $\sigma : I \to M$. We can pull this back through exp. Then we use Gauss Lemma *revisit and complete this computation!!!!* $\square$